

THE ECONOMIC EFFECT OF COVID LOCKDOWN POLICIES

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ABSTRACT. This paper presents a mathematical model of the COVID-19 pandemic. We extend the standard SIR model by including two new states, exposed (E) and death (D). We denote this model as the SEIRD model. We include additional transitions for vaccination, virus mutation, or waning immunity. We further augment the model dynamics with a function to model a COVID lockdown, parameterized by s and m . The parameter s controls the speed of a lockdown, and m controls the strength. We then present a total cost objective function that balances total deaths, economic cost, and a hospital over-capacity penalty. We follow by modifying the economic cost to account for individual-specific productivity levels. We present the optimal lockdown specification and conduct a sensitivity analysis for the most important parameters.

Keywords. Differential equations, COVID lockdown policy, mathematical modeling, epidemiology, optimization

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1. INTRODUCTION

Although the COVID-19 pandemic is behind us in the year 2025, it remains worthwhile to analyze to prepare for future pandemic events. For this paper, we focus on lockdown policies and examine the interplay between different policies and three cost measures: death, economic cost, and hospital overcapacity.

We make the simplifying assumption that a lockdown policy can be explained by two parameters: speed and strength. We allow policy to vary to capture the heterogeneity in speed and severity of lockdown policies across states. Some states operated via a gradual lockdown, starting with bans on large gatherings, moving to recommendations to work from home, then a stay-at-home mandate (e.g., Massachusetts). Others were more strict and swift, mandating a complete lockdown from day one (e.g., California).

The deterministic SIR model provides a strong baseline for modeling the COVID pandemic AlQadi and Bani-Yaghoub [4]. The model captures key characteristics, such as herd immunity, peak infection, and the dynamics of growth and slowdown in an epidemic. However, the SIR model is limited by three states: Susceptible (S), Infected (I), and Recovered (R). The parameters in the model are also restricted to be time-invariant, which may not be true, such as the case with transmission rate a . We build on the SIR framework to include two new states exposed (E) and death (D), and modify the transmission rate $a(t)$ to be time varying.

2. DESCRIPTION OF THE MODEL

The proposed SEIRD model (Beckley et al.) is specified by the following system of ordinary differential equations (ODE's):

$$\frac{dS}{dt} = -\frac{a(t)}{N}IS - cS + \gamma R \quad (2.1)$$

$$\frac{dE}{dt} = \frac{a(t)}{N}IS - \nu E \quad (2.2)$$

$$\frac{dI}{dt} = \nu E - bI - \eta I \quad (2.3)$$

$$\frac{dR}{dt} = bI + cS - \gamma R \quad (2.4)$$

$$\frac{dD}{dt} = \eta I. \quad (2.5)$$

The units are individuals and time t is measured in days. The five possible states for an individual are denoted as S , E , I , R , and D (Susceptible, Exposed, Infected, Recovered and Death, respectively). The inclusion of E was to model the latent period of COVID. When an individual is infected with COVID, they don't instantaneously become infectious. It takes time before they can transmit the virus to others. D is the terminal state of death, included to relax the SIR model assumption that all infected individuals recover.

In addition to the two new states, we include two new parameters c and γ , vaccination rate and mutation rate, respectively (Smith [22]). If a susceptible individual is vaccinated, they move to the recovered state. However, some recovered individuals go back to the susceptible state because, as the virus mutates, the initial vaccine becomes ineffective. The values for c and γ are fixed and were determined by Mathieu et al. [17] and Manica et al. [14]. The values given in the papers were converted to match our units. Figure 1 illustrates the graph depicting each state as a node and rates as edges.

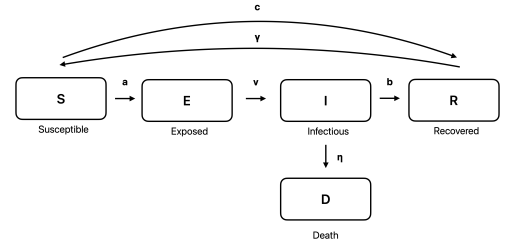


FIGURE 1. SEIRD Model Graph

The other parameters are the same as the SIR model. N denotes population. In our system, the population is closed (no births or deaths) fixed at $2e6$ to model a large city or MSA. b is the recovery rate from Oelsner et al. [18]. ν is incubation rate. η is the death rate given by the data from Mathieu et al. [17]. To convert from the case fatality rate (CFR) given in the data to the death rate used in our analysis, we use the formula and solve for η :

$$CFR = \frac{\eta}{\eta + b}. \quad (2.6)$$

For the entirety of this project, we assume the patient 0 scenario, starting with just one exposed individual. All parameter units are per person per day.

Finally, we modify a , the transmission rate, to be time-varying $a(t)$. The parameter can be decomposed as a product of the probability of transmission (pt) and the number of contacts, $nc(t)$:

$$a(t) = pt \cdot nc(t). \quad (2.7)$$

$nc(t)$ is time-dependent, modulated by the lockdown policy. The lockdown policy is defined as the following function:

$$nc(t) = m_{min} + (m_{max} - m_{min}) \cdot \exp[s \cdot \max(t - t_{start}, 0)]. \quad (2.8)$$

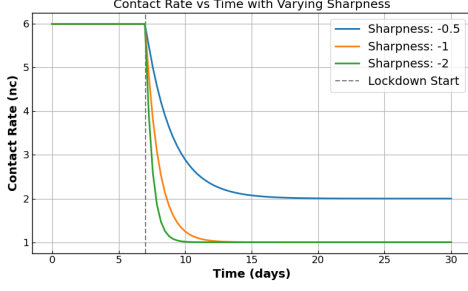
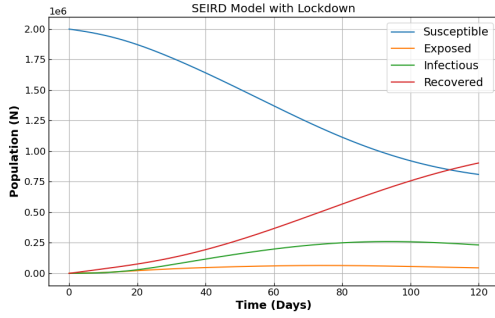


FIGURE 2. Lockdown Function

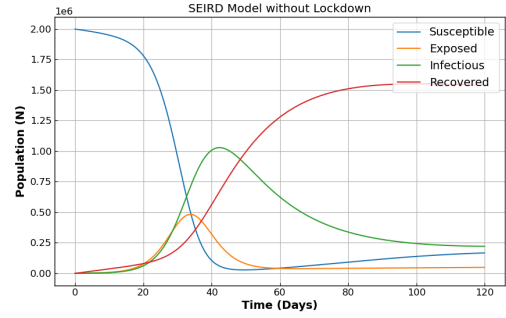
function takes on m_{max} . For $t \geq t_{start}$, the exponential decay begins.

For equation 2.7, we bound the number of contacts between 6 and 1. A lower bound of 1 makes sense because people at a minimum need to interact with one other person to survive (e.g., buy food, get water). A value of 6 is also reasonable because the average number of close friends is around 5 using Dunbar's number (Scotch [21]).

3. ANALYSIS



(A)



(B)

FIGURE 3. Solution Curves: Lockdown vs No Lockdown

In Figure 3, we present the solution curves for the system of ODEs, equations 2.1 - 2.5. Figure 3(a) shows the solution curves with a lockdown present, while Figure 3(b) presents solution curves in the absence of a lockdown. The lockdown has the effect of flattening the curves for the Infected and Exposed groups. We see the Susceptible group decline and the Recovered group increase because of vaccines. The peaks for Infected and Exposed are much more pronounced without lockdown. We see a lag between Exposed and Infected, a property of the SEIRD model. Cumulative death is omitted from the above figures for visual clarity, but can be found in Figure 4(B), displaying daily deaths for lockdown vs no lockdown.

The optimization problem is to minimize total cost by modifying decision variables s and m_{min} . Mathematically, we can express it as:

$$\begin{aligned} \min_{s, m_{min}} \quad & TC(s, m_{min}) \\ \text{s.t.} \quad & \end{aligned} \tag{3.1}$$

$$\begin{aligned} \text{SEIRD Model with Lockdown } nc(t, s, m_{min}) \\ s \leq 0, \end{aligned} \tag{3.2}$$

$$m_{min} \in [1, 5],$$

where:

$$TC(s, m_{min}) = V_{life} \cdot D(t) + GDP_{day} \int_0^T nc(t, s, m_{min}) dt + C_{hosp} \int_0^T \max(I(t) - H_{cap}) dt \quad (3.3)$$

The first term is the cost of deaths, which is the product $V_{life} \cdot D(t)$. V_{life} is the value of life in millions of dollars, and $D(t)$ is the cumulative number of deaths at the end of the period $[0, T]$. V_{life} was set to be \$15M but a sensitivity analysis on this parameter is conducted in Section 3.2. The DoT [8] finds it to be \$13.7M in 2024 or around \$14M in 2025 dollars. The second term is the economic cost. To obtain this value, we integrate the lockdown function $nc(t)$ over the $[0, T]$ to get the total lockdown days, and weigh it by the economic impact GDP_{day} . The implicit assumption is that lockdowns diminish productivity agnostic of the number of individuals in each SEIRD state. One can view the lockdown function as the fraction of GDP lost. This assumption will be relaxed in Section 3.1. Finally, we have the hospital overflow penalty. The integral gives the total number of overflows, and we multiply it by the cost per day per patient exceeding capacity. All cost terms are equally weighted, and units are in millions of dollars.

H_{cap} is a fixed parameter that is a proportion of N . We use data (“Fast Facts on U.S. Hospitals, 2025 — AHA”) to estimate the average hospital beds in the U.S., which we find to be around 0.3% of N . For C_{hosp} we use \$15k, derived from Kapinos et al. [11].

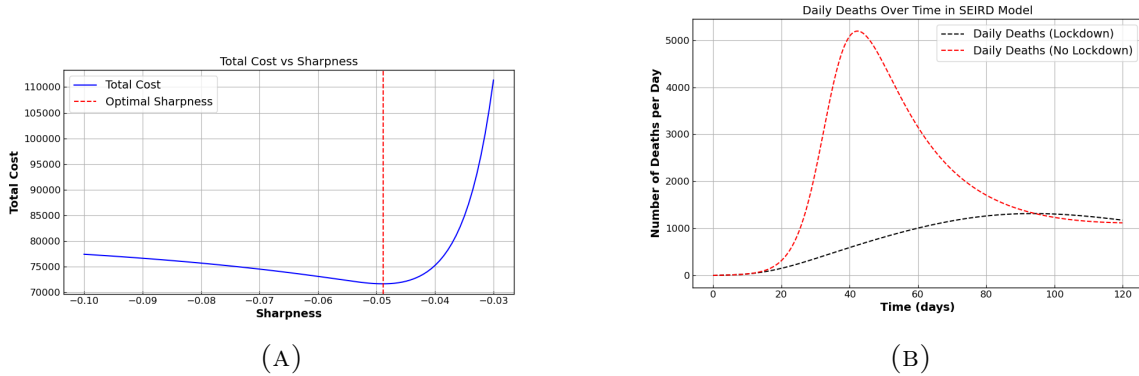


FIGURE 4. Optimal Sharpness and Daily Deaths

The optimum for the s is an interior point, as shown in Figure 4(A), while the optimum for m_{min} is on the boundary at 1. Many different parameterizations were attempted, and in all cases, m_{min} did not change. This may be due to the correlation of parameters or because it is always best to select the strongest lockdown. The latter makes sense because if one opts for a lockdown, it’s best to make it strong to eradicate the virus.

3.1. Cumulative Productivity Dependent on Lockdown Strategy. Now we relax the assumption that lockdowns diminish productivity regardless of the number of individuals in each SEIRD state. We adjust the model to consider the expected productivity values for each different population group in the SEIRD Model.

Calculating productivity is difficult to measure using one unit across different industries and individuals within an industry. To navigate this, we assumed a unit of productivity to be equal to the output generated by a healthy individual in a unit of time. In a fully healthy society, the number of units of productivity is equal to the number of people.

To use this with the defined SEIRD model, we will assign each population group in the model a productivity coefficient, resulting in the following total productivity calculation:

$$TP(t) = \sum_{i \in \{S, E, I, R, D\}} u_i \cdot P_i(t) \quad (3.4)$$

, where $TP(t)$ is a total productivity at a time t . u_i provides the productivity coefficient associated with the i state and $P_i(t)$ is the population of the i state at time t . Note that $i \in \{S, E, I, R, D\}$.

To evaluate different lockdown strategies, we evaluated three different scenarios: a sharp lockdown curve ($s = -0.1$), a no lockdown strategy, and a slow lockdown strategy ($s = -0.04$). All other variables in the SEIRD model were kept consistent across the three evaluations.

Both susceptible and recovered individuals are considered healthy and fully productive, so their productivity coefficient is 1. An infected person is considered a negative contribution to productivity, as they need care from other members in the population to recover. We set this coefficient to be -0.25, as we expect that a healthy individual could care for up to 4 infected individuals. This number is in line with Oregon House Bill 2697 [19], which provides registered nurse/patient limits for hospitals. We were unable to research an individual productivity decreases due to illness, so we will make an assumption. Based on the intuitive reason that a person progressing through an illness is less effective, we will assume a value of .8. Again, a deceased individual contributes no productivity.

For the no lockdown strategy, we set the array of productivity coefficients as follows.

$$u_i^{(\text{No Lockdown})} = \{1, .8, -0.25, 1, 0\} \quad (3.5)$$

In a lockdown scenario, these values change. Gibbs, Mengel, and Siemroth (2023) [9] observed that a transition to fully remote work results in a productivity between 8 to 19 percent. A 10 percent decrease appears reasonable with that context. With that change in productivity to susceptible and recovered individuals, the array of productivity coefficients is modified to:

$$u_i^{(\text{Lockdown})} = \{.9, .8, -0.25, .9, 0\} \quad (3.6)$$

This lockdown productivity coefficients array will be the same for both a sharp and a slow lockdown. The population counts will differ between the two total productivity calculations, leading to a difference in the total productivity observed.

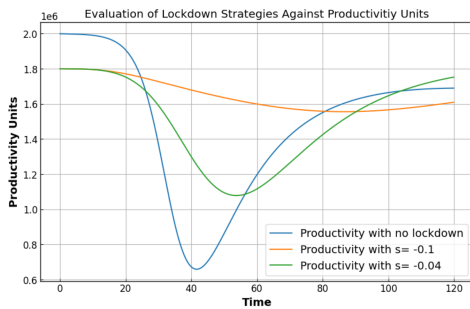


FIGURE 5. Productivity as a Result of Different Lockdown Strategies

To understand final productivity levels, we assumed that once there have been no infections for 14 days, the lockdown is lifted and all healthy populations return to full effectiveness. This period is based on the common isolation period defined by most governments during the pandemic, such as the U.S. Department of State's Recommendation in March 2020 [23]. We then calculated the total productivity using these productivity coefficients and the population dynamics from the SEIRD model using a slow, sharp, and no lockdown strategy. The results can be seen in Figure 5.

This chart follows intuition in that for the post-epidemic steady state, the lockdown strategies will result in higher productivity. This results from having fewer deaths during the pandemic, and therefore, more individuals can contribute. A surprising observation is that during the pandemic, the no-lockdown

strategy ended up having the lowest productivity. Intuition would state that with more people working normally, productivity would exceed strategies incorporating a lockdown. However, this analysis suggests that the cost to treat infected individuals outweighs the benefit of people working in person.

3.2. Sensitivity Analysis. We now consider the sensitivity of our result to changes in three parameters: death rate (η), probability of transmission (pt), and GDP per Day (GDP_{day}). We first examine the sensitivity of the optimal lockdown speed with respect to changes in GDP_{day} , the weight on economic cost. The higher the GDP_{day} , the more costly the lockdown in economic terms. Next, we examine the sensitivity of pt to the optimal lockdown speed. It is important to analyze this parameter because the probability of transmission varies by environment. For households, it ranges from 0.10-0.40. For shared indoor spaces, it ranges from 0.01-0.05 (Bulfone et al. [7]). Varying this parameter allows us to see how optimal lockdown speed varies between seasons. In winter seasons, pt is higher since most people are interacting indoors, while in summer it is the opposite.

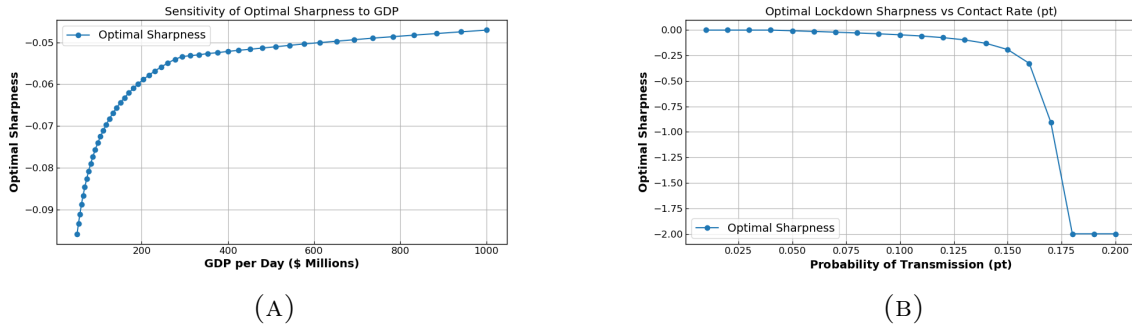


FIGURE 6. Sensitivity of Optimal Lockdown Speed to pt and α

Figure 6(a) shows the sensitivity of GDP_{day} . The range of values chosen for the sensitivity analysis of GDP_{day} was based on the GDP per capita averages across different regions of the U.S.. The conversion between GDP per day and GDP per capita is given by $GDP \text{ per capita} = \frac{GDP \text{ per day} * 365}{N}$.

At the lowest range, we have GDP per day of 50 million, which translates to about \$9k GDP per capita, close to many rural areas in the U.S.. At the upper range, we have \$182k GDP per capita, which is close to high productivity areas like Silicon Valley and parts of New York City (Wikipedia contributors [25]). In Figure 4(a), we see that as GDP_{day} increases, the optimal lockdown speed slows. This makes intuitive sense because as one values the economy more, lockdowns become less favorable. The shape of the curve is related to the curvature of the lockdown function and does not necessarily reflect diminishing returns.

Turning our attention to Figure 6(b), we see that for low values of the parameter pt , the optimal speed is very slow. For a low probability of transmission, a quick lockdown isn't necessary. As pt increases, the lockdowns become quicker. The optimal s takes on

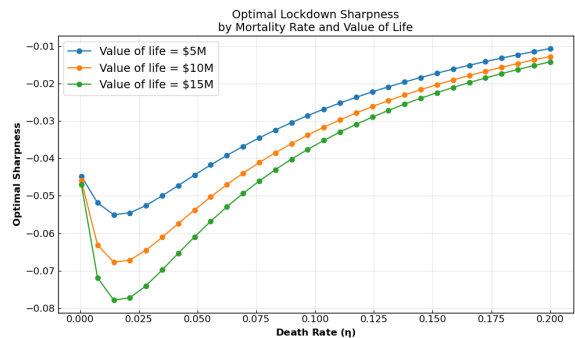


FIGURE 7. Sensitivity of Optimal Lockdown Speed to Death Rate (η)

the maximum value on the boundary at -2. This is again intuitive because if a virus is spreading quickly, it's best to enact a lockdown swiftly.

Finally, we present a more interesting case, which is the sensitivity of optimal lockdown speed to death rate η . From Figure 6, the relationship between death rate and the optimal lockdown speed is not monotonic, as it was for the prior two parameters. The intuition is that for low death rates, a swift lockdown is preferred. However, as the death rate increases, a lockdown can only do so much to prevent deaths; therefore, the objective function starts to prefer the economy over lives when pt is large. The curves in Figure 6 also vary by the value of life. As the value of life increases, the objective function prefers sharper lockdowns, matching intuition.

4. DISCUSSION

For our analyses, we fixed the initial condition to start with a single Exposed individual, a patient zero scenario. We argue that starting with this initial condition is the most realistic because a location can't immediately spawn 10 exposed individuals unless there is traveling between locations, but the model does not take this into account. It is meant to model a single environment. N is fixed at $2e6$; changing the value doesn't change the outcome. We also acknowledge that the value of a human life is subjective, hence the sensitivity analysis, which allows it to range from \$5M to \$15M.

An exploration of adding additional parameters was conducted, such as the length of the lockdown and allowing the lockdown start time to vary. However, the length of the lockdown was found to be collinear with the speed of the lockdown. This makes sense given that a slower lockdown implies a longer, dragged-out lockdown period. Varying the start of the lockdown didn't yield interesting results because the optimizer always prefers to choose the earliest lockdown time.

Most results are intuitive; however, the sensitivity analysis reveals an interesting result for the death rate. Lockdowns are effective only up to a point. When death rates are very high, lockdowns become ineffective. In our simple objective function, it would be better to allow unavoidable deaths in favor of saving the economy.

Overall, the model achieved what we hoped to accomplish. It allowed us to model vaccinations, mutations, and the incubation period. The model also allowed us to determine different optimal lockdown policies based on the tradeoffs between death, economic cost, and hospital capacity. We even went further to model state-specific productivity to see how it changes during a lockdown.

5. CONCLUSION

Fortunately, the days of the pandemic are over, but it is still worthwhile to analyze it to learn from prior mistakes. The model reveals a great deal about the dynamics between lockdown policies and SEIRD dynamics. In this project, we extended the SIR to include new states to better capture the nuances of a virus. We then presented a functional form to model a lockdown policy, parameterized by the speed and strength of a lockdown. We followed with an objective function that captures the total cost of a lockdown, extending it to analyze state-specific productivity during a lockdown. The sensitivity analysis revealed the limitations of a lockdown. If death rates are too high, there is only so much a lockdown can do. We acknowledge that our model doesn't capture the complete picture of the pandemic. Modifications to include factors such as age-dependent mortality rates or conducting additional sensitivity analysis on parameters are all viable extensions to the model.

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6. STATEMENT OF INDIVIDUAL CONTRIBUTION

Kaylee

- Worked on the SEIRD model, the optimization problem and sensitivity analysis.
- Wrote intro, description of the model, the first section of the analysis, discussion and conclusion.

Tadhg

- Worked on the economic productivity model
- Wrote the second section of the analysis
- Built the PowerPoint